Your agent isn't guessing

You just didn't decide yet

Spec-Driven Development for coding with AI agents

Dell APJ AI Innovation Hub

DELL Technologies



TDD, BDD, SDD

TEST-DRIVEN DEVELOPMENT

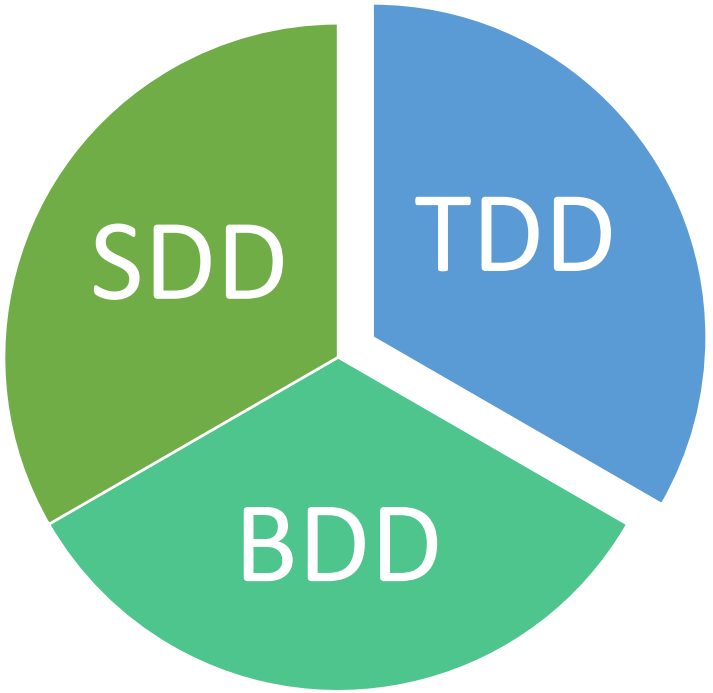
- Write a **test first**, then write code to make the test pass
- How you verify one behavior

BEHAVIOR-DRIVEN DEVELOPMENT

- Write **behavior/scenario first** in business language, then implement against those scenarios
- How users or the business expect it to act

SPEC-DRIVEN DEVELOPMENT

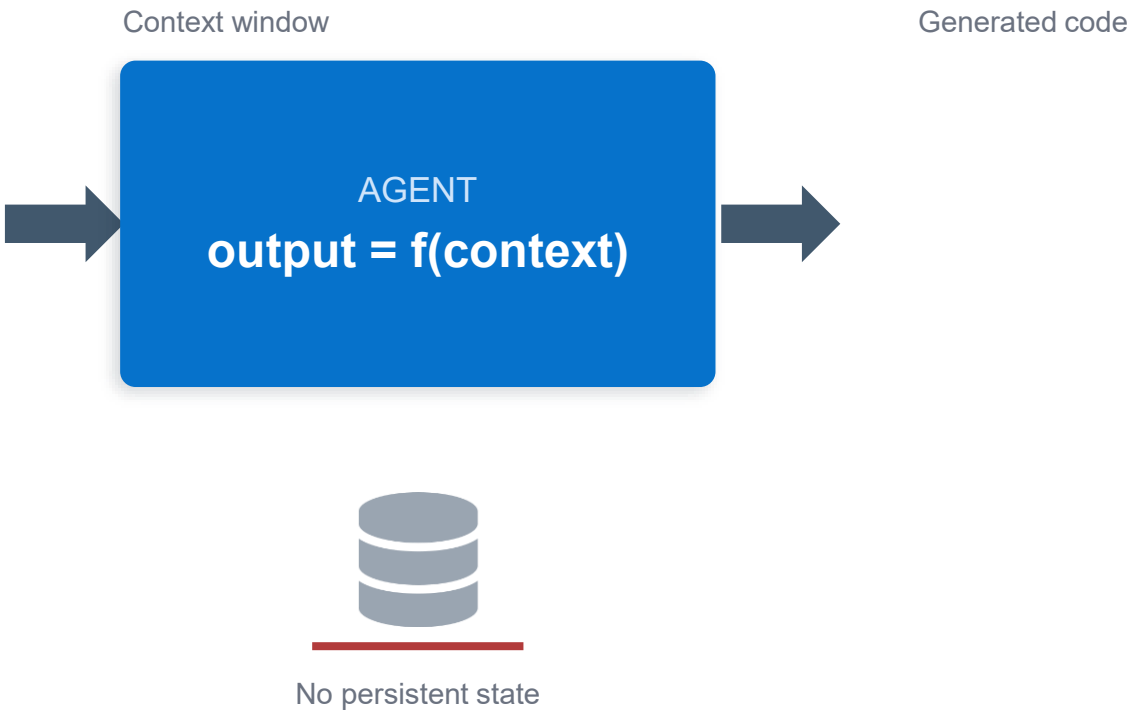
- Write the **spec first**, then implement to match it
- What the system should do



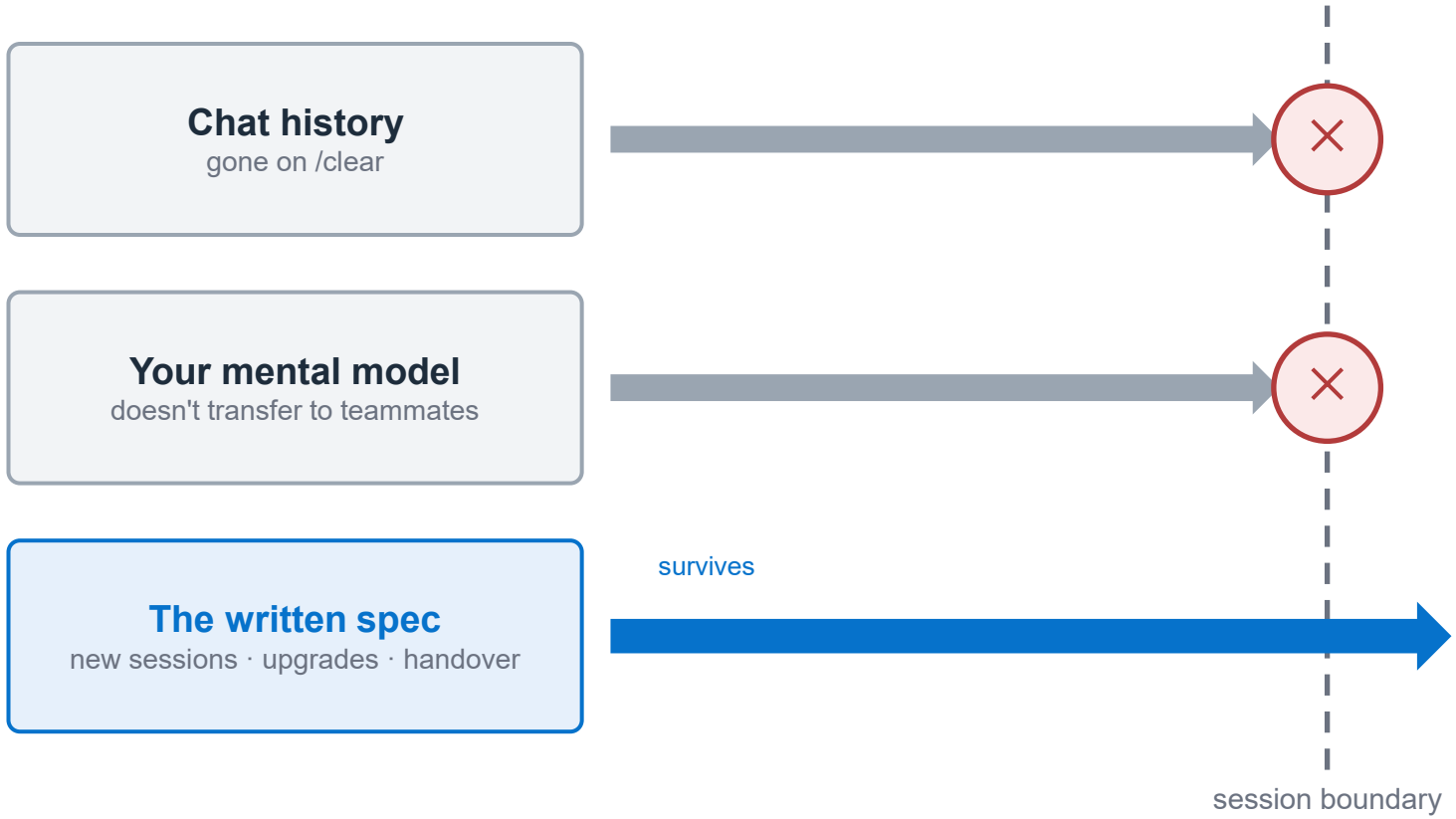
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The agent is a function of its context window, not of your cleverness

- No memory of your intent — only the text in front of it
- What you know but haven't written down is invisible
- Output quality is bounded by the context you assembled
- "Prompt harder" optimises the part that gets thrown away



Only one part of your context survives the session



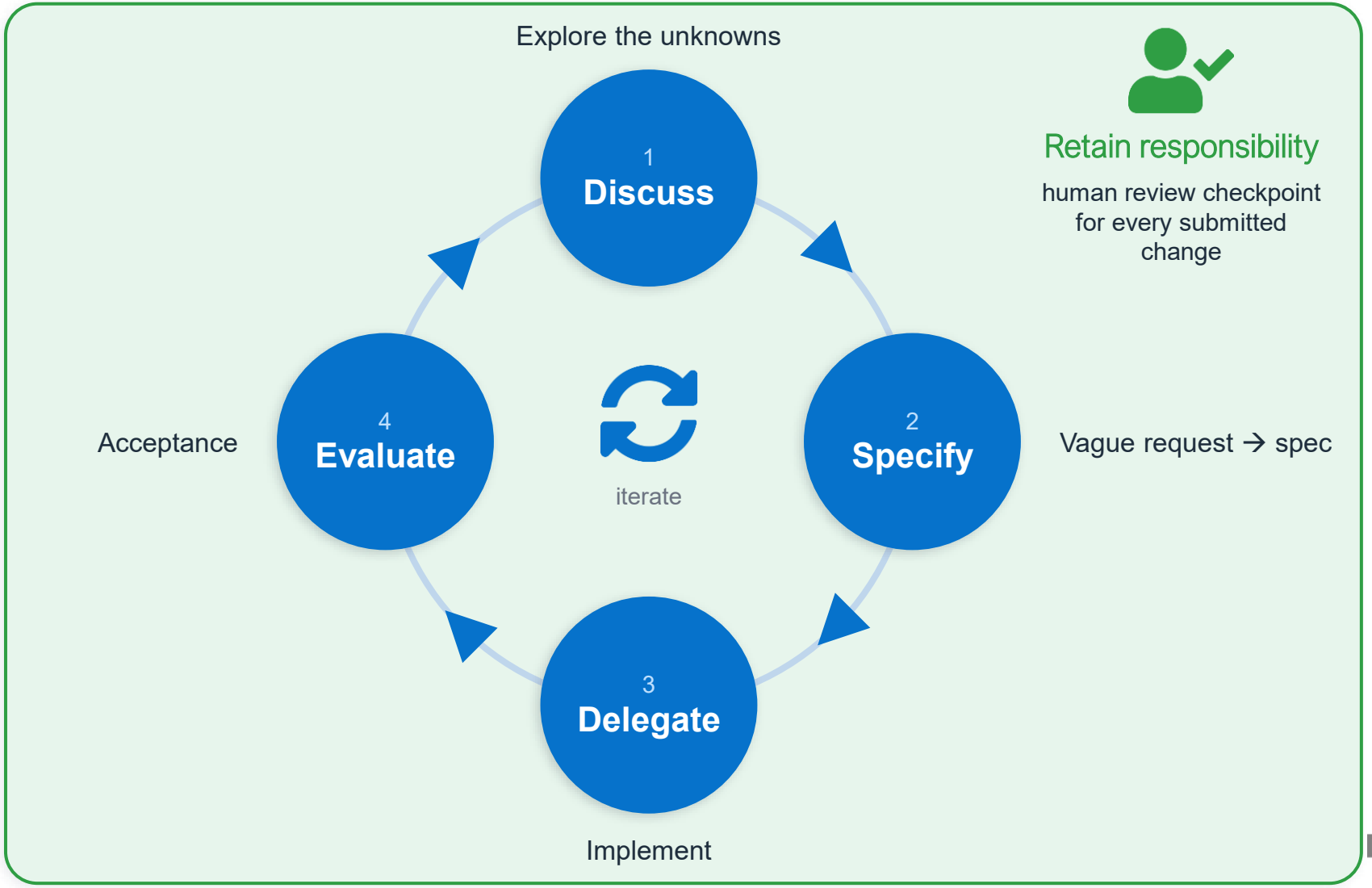
We write specs for durability — not for the agent.

Goal Is to Make AI-Assisted Work Inspectable: Discuss, Specify, Delegate, Evaluate

VIBE CODING

Vague request → Plausible code → Hope

✗ *fragile & ungrounded*



A Plausible Answer Can Still Solve the Wrong Problem

THE PROMPT

"Add a login feature to our app."

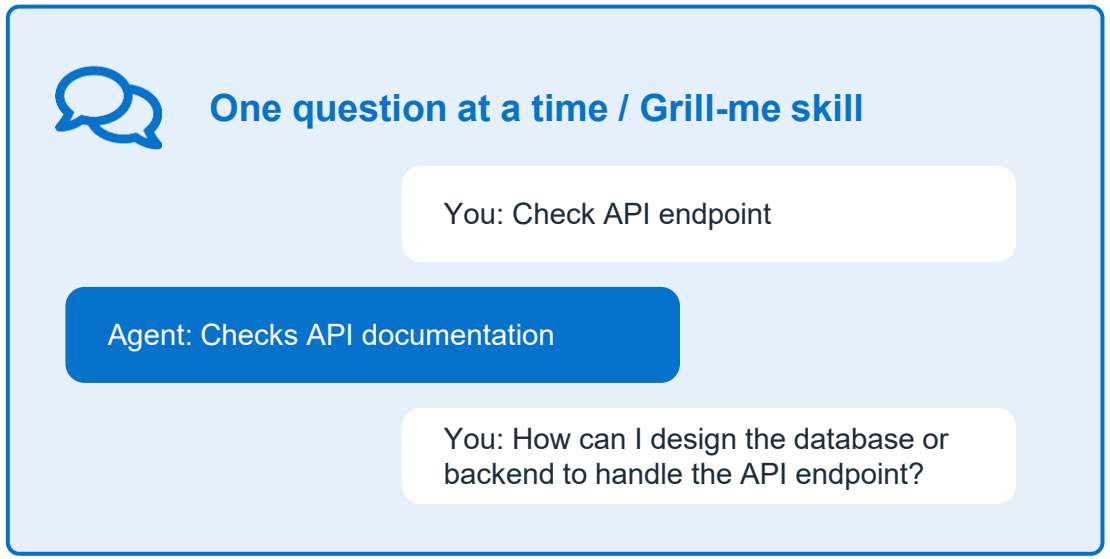
Longer prompts don't fix missing engineering decisions.

- Which users can register or sign in?
- What authentication method is permitted?
- What happens after failed attempts?
- Which security and privacy constraints apply?
- How do we know the feature is complete?

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Discuss: make the decisions before the agent makes them for you

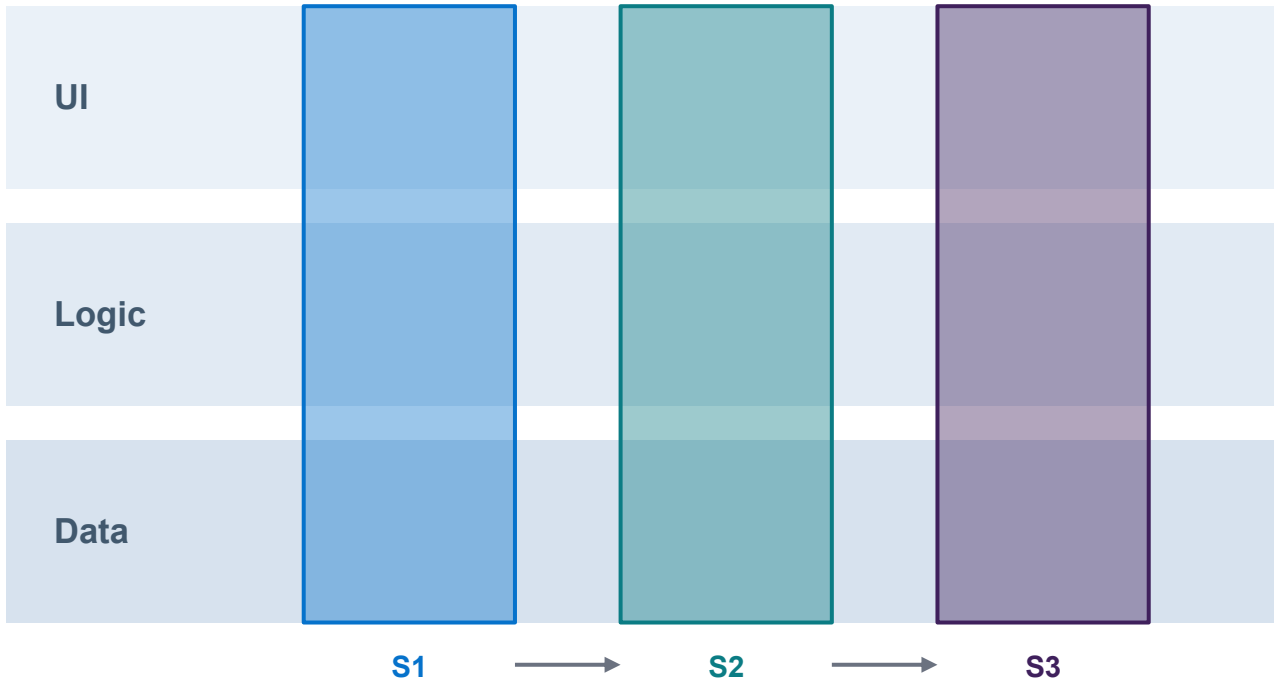
- Research external sources and the existing codebase
- Prototype design options; state assumptions out loud
- Name what is explicitly out of scope



Discussion questions	
Goal	What user outcome are we trying to achieve?
Constraints	What boundaries and project conventions must remain?
Non-goals	What related work is explicitly excluded?
Acceptance criteria	*e.g.* **When Button is selected, only tasks whose completed value is false are displayed.**
Checks	Which test cases should be assessed for this feature?

Specify produces the PRD; Delegate implements it

- Spec = minutes of the meeting you had with the agent, for example via /to-spec skill
- Goal, user, constraints, criteria, functional requirements
- Break down the tasks into smaller issues, for example via /prd-to-plan skill



- Sets up manageable issues for agents to work on separately, focused context window
- Has a clear test
- Manage dependencies between them

During implementation, use feedback loops to prevent errors compounding

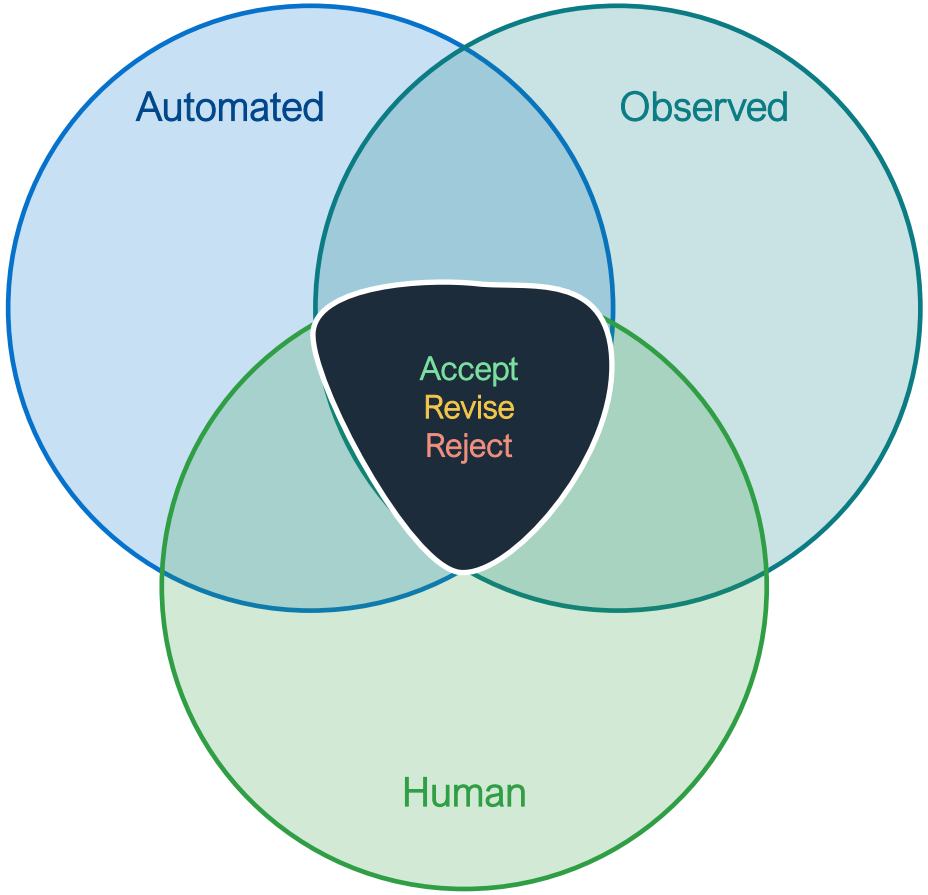
Feedback loop	Prevents
Typechecks	Mismatched types, wrong function shapes
Unit tests	Broken logic, silent regressions
Linter rules	Sloppy style, likely bug patterns
Pre-commit hooks	Careless work reaching the shared branch



self-correction loop

Without them, errors compound across iterations.

Evaluation



Automated evidence

- Unit & integration tests
- Type checks, lint, builds
- Security & dependency checks

Observed evidence

- Runtime & UI behavior
- Logs & failure reproduction

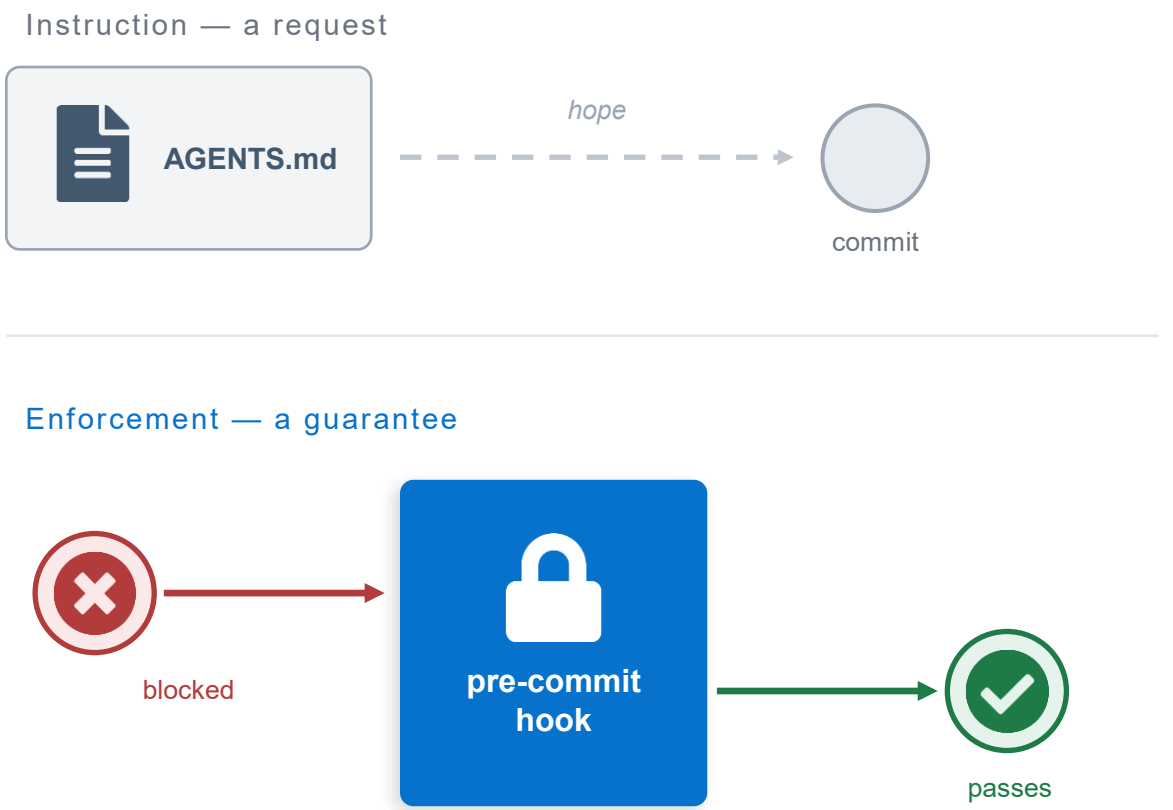
Human evidence

- Requirement traceability & diff review
- Security, privacy, maintainability

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You can't instruct reliability into an agent — you have to enforce it

- "Run the tests" in AGENTS.md or the prompt does not guarantee compliance
- LLMs are probabilistic — they don't execute instructions like code does
- As context fills, earlier instructions get lost or outweighed
- This is how the models work — so enforce it with a pre-commit hook
- Agents make 80-100% code coverage tests achievable



Stop copy-pasting errors into the chat — give the agent the logs

- Without log access you are the courier — every error pasted in by hand
- Automate it! With access, the agent diagnoses, fixes and re-tests on its own
- Structured logs beat raw strings — raw files get large and hard to parse
- Filter by source, level or failed HTTP status, msg, err.name and status code make agent search possible → names the log file and the specific incident to investigate

